SECTION 1: Electric Charges & Properties (Q1–10)

1. Which of the following is a property of electric charge?
Answer: B) It is conserved
Explanation: Electric charge is **quantized** and **conserved**, meaning it cannot be created or destroyed, only transferred from one body to another.

2. The SI unit of electric charge is:
Answer: B) Coulomb
Explanation: The SI unit of charge is the **Coulomb (C)**. 1 C = charge transported by 1 ampere in 1 second.

3. Two bodies attract each other. The bodies are:
Answer: C) One positive, one negative
Explanation: Opposite charges attract; like charges repel.

4. Which of the following is always true for an isolated system?
Answer: B) Total charge remains constant
Explanation: In an isolated system, **net electric charge is conserved**, though charges can move between objects.

5. What happens when a rubber rod is rubbed with fur?
Answer: A) Rubber gains electrons
Explanation: Rubber is an electron **acceptor**, and fur loses electrons, giving the rod a **negative charge**.

6. Which instrument is used to detect electric charge?
Answer: C) Electroscope
Explanation: An electroscope detects charge through **leaf divergence** when charged.

7. Which of the following is a conductor?
Answer: B) Copper
Explanation: Metals like copper have **free electrons**, so they conduct electricity.

8. Which charge is carried by an electron?
Answer: B) Negative
Explanation: Electrons carry **negative charge**, protons carry positive.

9. The principle of conservation of charge was first stated by:
Answer: B) Benjamin Franklin
Explanation: Franklin proposed the idea that **total charge is conserved**.

10. Which of the following statements is correct?
Answer: C) Net charge in an isolated system is conserved
Explanation: Charges cannot appear or disappear spontaneously; they can only move or redistribute.

SECTION 2: Conductors, Insulators & Induction (Q11–20)

11. Which of the following is an insulator?
Answer: C) Glass
Explanation: Glass **does not allow free movement of electrons**, hence it's an insulator.

12. Charging by friction occurs due to:
Answer: A) Electron transfer
Explanation: Friction transfers electrons from one object to another.

13. In charging by induction, the object:
Answer: B) Does not touch the charged body
Explanation: Induction charges an object **without contact**, via redistribution of electrons.

14. The electric field inside a conductor in electrostatic equilibrium is:
Answer: B) Zero
Explanation: In electrostatic equilibrium, **electrons redistribute** so the internal field cancels.

15. A hollow metallic sphere is charged. The charge resides:
Answer: B) On outer surface
Explanation: Charges on a conductor move to the **outer surface** to minimize repulsion.

16. Earthing is done to:
Answer: B) Prevent charge accumulation
Explanation: Earthing provides a **path for excess charge** to flow to the Earth.

17. Which of the following statements is true?
Answer: C) Conductors allow electron movement
Explanation: Conductors have **free electrons** that can move easily.

18. A charged comb attracts small paper pieces due to:
Answer: B) Polarization
Explanation: Neutral paper becomes polarized; opposite charges attract the comb.

19. Why does a sharp point on a conductor leak charge easily?
Answer: B) High potential gradient
Explanation: At sharp points, **electric field is stronger**, so charge leaks easily.

20. A material which partially conducts electricity is:
Answer: C) Semiconductor
Explanation: Semiconductors have **moderate conductivity** between conductors and insulators.

SECTION 3: Coulomb's Law (Q21–30)

21. Coulomb's law states that force between two charges is:
Answer: B) Inversely proportional to square of distance
Explanation: $F = k\frac{q_1 q_2}{r^2}$

22. The electrostatic force is:
Answer: C) Attractive or repulsive
Explanation: Like charges repel, opposite charges attract.

23. If the distance between two charges is doubled, the force becomes:
Answer: C) 1/4
Explanation: $F \propto \frac{1}{r^2}$, so r doubled → F becomes 1/4.

24. The SI unit of Coulomb's constant is:
Answer: A) $N \cdot m^2/C^2$
Explanation: In $F = k\frac{q_1 q_2}{r^2}$, k has units **$N \cdot m^2/C^2$**.

25. Coulomb's law is valid:
Answer: B) Only for point charges
Explanation: Valid strictly for **point charges** or **spherically symmetric charge distributions**.

26. Two equal positive charges are placed 1 m apart. Force is F. If one charge is doubled, force becomes:
Answer: B) 2F
Explanation: $F \propto q_1 q_2$, doubling one charge → force doubles.

27. Electrostatic force is a:
Answer: A) Vector
Explanation: Force has **magnitude and direction**.

28. The principle of superposition is used to:
Answer: A) Calculate net force
Explanation: Forces from multiple charges **add vectorially**.

29. Which of these affects Coulomb force?
Answer: D) All of the above
Explanation: Magnitude of charges, distance, and medium all affect the force.

30. Two point charges attract each other with 36 N. If each charge is halved, the new force is:
Answer: A) 9 N
Explanation: $F \propto q_1 q_2$, halving both → F reduces to $\frac{1}{4} \times 36 = 9$ N.

SECTION 4: Electric Field & Field Lines (Q31–50)

31. Electric field at a point is:
Answer: A) Force per unit charge
Explanation: $E = F/q$

32. SI unit of electric field is:
Answer: A) N/C
Explanation: Force per unit charge → N/C

33. Electric field lines start from:
Answer: B) Positive charges
Explanation: Field lines originate from positive and terminate at negative charges.

34. Electric field lines end at:
Answer: B) Negative charges
Explanation: Opposite charges attract field lines terminate at negative.

35. Which is true about field lines?
Answer: B) They never intersect
Explanation: Two field lines cannot cross; direction would be ambiguous.

36. Field inside a uniformly charged spherical shell is:
Answer: B) Zero
Explanation: By Gauss's Law, **inside a shell, field = 0**.

37. Electric field due to point charge decreases as:
Answer: B) $1/r^2$
Explanation: $E = kq/r^2$

38. A test charge is:
Answer: B) Very small
Explanation: A test charge is small enough **not to disturb the field**.

39. Field lines for dipole:
Answer: C) Curved
Explanation: Dipole field lines are **curved from positive to negative**.

40. Uniform electric field is produced between:
Answer: A) Parallel plates
Explanation: Parallel plates with equal and opposite charges produce **uniform field**.

41. Tangent to electric field lines gives:
Answer: A) Direction of field
Explanation: Tangent at a point indicates **field vector direction**.

42. The electric field is strongest:
Answer: A) Where lines are close
Explanation: Density of lines $\propto$ field strength.

43. For a negative point charge, field lines:
Answer: B) Point inward
Explanation: Field lines **point toward negative charges**.

44. Electric field inside a conductor:
Answer: A) Zero
Explanation: Free electrons redistribute to cancel internal field.

45. Electric field due to infinite line charge is proportional to:
Answer: B) 1/r
Explanation: From Gauss's law, $E = \frac{\lambda}{2\pi\epsilon_0 r}$

46. Which of these is a vector quantity?
Answer: B) Electric field
Explanation: Electric field has **magnitude and direction**.

47. Neutral point in dipole:
Answer: A) E = 0
Explanation: Net field = 0 at neutral points along the axis.

48. Electric field at equatorial line of dipole is:
Answer: B) Perpendicular to dipole axis
Explanation: Field at equator is **perpendicular to the dipole axis**.

49. Electric flux is:
Answer: A) Scalar
Explanation: Electric flux $\Phi = E \cdot A$ is a **scalar** quantity.

50. The SI unit of electric flux is:
Answer: A) N·m²/C
Explanation: Flux units = $E \times area = N/C \cdot m^2 = N \cdot m^2/C$